

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

**Annexure A
Comparative Analysis**

Code of Conduct:

I D.Uday Kiran(192425161) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D.Uday Kiran

Question 1: Compare Star Topology and Bus Topology in terms of Cost, Reliability, and Performance

Understanding of Concepts

Star topology is a network topology in which every device is connected individually to a central device such as a switch or hub. All communication between devices generally passes through this central device.

Bus topology uses a single backbone cable to which all devices are connected. Devices share the same communication medium for transmitting data.

The major differences between these two topologies can be analyzed using **cost, reliability, performance, scalability, and maintenance**.

Comparison Depth

Cost:

Bus topology is generally less expensive because it uses a single backbone cable and requires fewer cables and networking devices. Star topology requires individual cables from every device to the central switch or hub. Therefore, the initial installation cost of star topology is higher.

Reliability:

Star topology provides better reliability. If one computer or its connecting cable fails, the remaining computers can continue communicating. In bus topology, failure of the main backbone cable can affect the entire network.

Performance:

Star topology normally provides better performance because a switch can manage communication efficiently. In bus topology, all devices share the same communication medium. As more devices are added, network traffic and collisions can increase.

Scalability:

Star topology is easier to expand because new devices can be connected to available ports on the switch. Bus topology becomes difficult to expand as the number of devices increases because additional devices increase traffic on the shared backbone.

Maintenance:

Star topology is easier to troubleshoot because each device has an individual connection. In bus topology, identifying a fault in the common backbone can be difficult.

Criteria Selection

The most relevant criteria for comparing star and bus topology are:

- Installation cost

- Reliability
- Network performance
- Scalability
- Maintenance
- Fault tolerance
- Ease of troubleshooting

These criteria directly determine the suitability of a topology for a particular network environment.

Analytical Skills

Although bus topology is cheaper, its shared communication medium creates limitations. When the number of devices increases, traffic also increases, which can reduce performance.

Star topology has a higher initial cost because of the central switch and individual cables. However, this additional cost provides better performance, easier fault detection, and greater reliability.

For example, if one computer fails in a star network, only that computer is normally affected. If the backbone cable fails in a bus network, communication for the entire network can be interrupted.

Use of Examples / Evidence

Consider a network with **10 computers**.

In a star topology, each computer is connected separately to a switch. If Computer 5 fails, Computers 1–4 and 6–10 can continue communicating.

In a bus topology, all 10 computers share the same backbone. If the backbone cable is damaged, all computers may lose network connectivity.

Therefore, for a modern office or computer laboratory, star topology is generally more suitable.

Clarity of Presentation

Factor	Star Topology	Bus Topology
Cost	Higher	Lower
Cabling	More	Less
Reliability	High	Lower
Performance	Better	Decreases with traffic
Scalability	Good	Limited
Maintenance	Easy	More difficult

Factor	Star Topology	Bus Topology
Fault Isolation	Easy	Difficult

Question 2: Compare Mesh Topology and Star Topology in terms of Fault Tolerance and Reliability

Understanding of Concepts

Mesh topology connects devices using multiple communication links. In a full mesh topology, every device is directly connected to every other device.

Star topology connects all devices to a central switch or hub.

Fault tolerance refers to the ability of a network to continue operating even when one or more components fail.

Comparison Depth

Fault Tolerance:

Mesh topology has very high fault tolerance because multiple communication paths are available. If one link fails, data can travel through another path.

Star topology has moderate fault tolerance. Failure of an individual device normally does not affect other devices. However, failure of the central switch or hub can affect the entire network.

Reliability:

Mesh topology is highly reliable because of its redundant connections. Star topology is also reliable, but its central device represents a possible single point of failure.

Cost:

Mesh topology is expensive because it requires many cables and ports. Star topology requires fewer connections and is therefore more economical.

Complexity:

Mesh topology is more complex to install and maintain. Star topology is comparatively simple.

Criteria Selection

Important criteria include:

- Fault tolerance
- Reliability
- Redundancy
- Cost
- Installation complexity
- Maintenance

- Failure impact

Analytical Skills

The major strength of mesh topology is **redundancy**. Multiple paths are available between devices, so the network can continue operating even when a link fails.

For a full mesh containing n devices:

$$\text{Number of links} = n(n - 1) / 2$$

For 5 devices:

$$5(5 - 1) / 2 = 10 \text{ links}$$

Therefore, 5 devices require 10 direct connections in a full mesh.

In star topology, the same 5 devices require only 5 connections to the central switch.

Thus, mesh provides better fault tolerance but at a significantly higher cost and complexity.

Use of Examples / Evidence

Consider a company where continuous communication is required.

If a connection between two devices in a mesh network fails, another path can be used. Communication can continue.

In a star network, if one workstation fails, the remaining workstations continue working. However, if the central switch fails, communication between all connected devices can be interrupted.

Therefore, mesh topology is suitable for highly critical environments where network availability is essential.

Clarity of Presentation

Factor	Mesh Topology	Star Topology
Fault Tolerance	Very High	Moderate
Reliability	Very High	High
Redundancy	Very High	Low/Moderate
Cost	Very High	Moderate
Installation	Complex	Simple
Maintenance	Difficult	Easy

Factor	Mesh Topology	Star Topology
Central Failure	No single central point	Can affect entire network

Question 3: Compare Bus Topology and Mesh Topology in terms of Installation Complexity

Understanding of Concepts

Bus topology uses a **single backbone cable** to connect multiple devices. It has a simple physical structure and requires relatively few connections.

Mesh topology uses multiple links between devices. In a full mesh, every device is directly connected to every other device.

Therefore, the installation complexity of mesh topology is much higher than that of bus topology.

Comparison Depth

Bus Topology:

- Requires a single backbone cable.
- Uses fewer cables.
- Requires fewer network connections.
- Easier to install.
- Less expensive.
- Suitable for small networks.

Mesh Topology:

- Requires multiple connections.
- Requires more cables.
- Requires more ports.
- Takes longer to install.
- Costs more.
- Requires careful planning.

The number of links in a full mesh is:

$$\text{Number of links} = n(n - 1) / 2$$

Criteria Selection

The comparison can be made using:

- Number of cables

- Number of connections
- Installation time
- Cost
- Configuration complexity
- Scalability
- Maintenance requirements

Analytical Skills

Bus topology has a simple structure because devices share one backbone cable. This reduces installation requirements.

However, mesh topology becomes increasingly complex as devices are added.

For example:

Number of Devices Mesh Links Required

4	6
5	10
10	45
20	190

This shows that mesh complexity increases rapidly.

Although mesh topology is difficult to install, the additional connections provide redundancy. Therefore, its complexity provides an important benefit in terms of reliability.

Use of Examples / Evidence

Suppose a small laboratory has 5 computers.

A bus network can connect the computers through one backbone cable.

A full mesh network would require:

$$5(5 - 1) / 2 = 10 \text{ links}$$

Therefore, mesh requires significantly more physical connections.

A small temporary network may therefore prefer bus topology, while a critical network may choose mesh topology because reliability is more important than installation simplicity.

Clarity of Presentation

Factor	Bus	Mesh
Installation	Easy	Difficult
Cabling	Low	Very High
Cost	Low	High
Configuration	Simple	Complex
Installation Time	Short	Long
Scalability	Limited	Complex
Reliability	Lower	Very High

Question 4: Compare Hybrid Topology and Star Topology in terms of Scalability

Understanding of Concepts

Star topology connects devices to a central switch or hub. It can be expanded by connecting additional devices to unused ports.

Hybrid topology combines two or more different topologies, such as star, bus, ring, or mesh. This allows an organization to design different network sections according to specific requirements.

Scalability refers to how easily a network can grow when new devices, users, departments, or locations are added.

Comparison Depth

Star topology provides good scalability for small and medium-sized networks. New devices can be added if the central switch has available ports.

However, the central device has a limited number of ports. Once all ports are occupied, another switch or networking device must be added.

Hybrid topology provides greater scalability because different network segments can be added independently.

For example, a university can have:

- Star topology in the Computer Science department
- Star topology in the Mechanical department
- Mesh connections for critical systems
- Different network segments connected together

This creates a larger hybrid network.

Criteria Selection

Relevant comparison criteria include:

- Expansion capability
- Network size
- Flexibility
- Cost
- Port limitations
- Network segmentation
- Management complexity
- Future growth

Analytical Skills

The main limitation of star topology is its dependence on the central device.

For example, if a switch has 24 ports and all 24 ports are occupied, adding more computers requires additional networking hardware.

Hybrid topology avoids this limitation by allowing separate network segments to be connected.

This makes hybrid topology more suitable for large organizations such as universities, hospitals, multinational companies, and large industries.

However, hybrid topology is more complex to design and maintain because different topologies may have different requirements.

Use of Examples / Evidence

Consider a university with several departments.

A single star network may become difficult to manage when thousands of computers are connected.

Instead, each department can have its own star network, and these departmental networks can be interconnected. This produces a hybrid network.

If a new department is created, a new network segment can be added without completely redesigning the existing network.

Clarity of Presentation

Factor	Star Topology	Hybrid Topology
Scalability	Good	Very High
Flexibility	Moderate	Very High
Expansion	Depends on switch ports	Easy through new segments

Factor	Star Topology	Hybrid Topology
Cost	Moderate	Higher
Complexity	Low	High
Maintenance	Easy	Moderate/Complex
Large Networks	Suitable	Highly Suitable

Question 5: Compare Star, Mesh, Bus, and Hybrid Topologies in terms of Maintenance

Understanding of Concepts

Network maintenance involves monitoring the network, detecting faults, troubleshooting connectivity problems, replacing faulty components, and ensuring reliable communication.

The maintenance requirements of star, mesh, bus, and hybrid topologies are different because each topology has a different physical and logical structure.

Comparison Depth

Star Topology:

Star topology is generally easy to maintain. Each device has a separate connection to the central switch. If one device fails, the administrator can inspect that particular connection without necessarily affecting other devices.

Mesh Topology:

Mesh topology is difficult to maintain because it contains a large number of connections. Although the network is highly reliable, troubleshooting multiple links can take considerable time.

Bus Topology:

Bus topology is simple in structure, but troubleshooting can be difficult when the backbone cable fails. Since many devices share the same cable, a backbone problem can affect the entire network.

Hybrid Topology:

Hybrid topology has moderate to high maintenance complexity because it combines different topologies. Administrators must understand and manage different network segments and networking equipment.

Criteria Selection

The most relevant maintenance criteria are:

- Fault detection
- Troubleshooting difficulty
- Number of connections

- Failure impact
- Maintenance cost
- Technical expertise
- Network monitoring
- Ease of component replacement

Analytical Skills

Star topology generally provides the best balance between reliability and maintenance.

If one computer fails, the administrator can isolate the problem to that computer or its cable.

In mesh topology, multiple paths increase reliability but also increase the number of components that need to be monitored.

Bus topology has fewer components, but the backbone cable is critical. A single backbone failure can affect many devices.

Hybrid topology can provide better segmentation. If properly designed, a problem in one network segment may be isolated from other segments. However, maintaining different types of topologies requires skilled administrators.

Use of Examples / Evidence

Consider an office with 20 computers.

In a star network, each computer is connected to a central switch. If Computer 12 loses connectivity, the administrator can inspect its cable and switch port.

In a bus network, if all 20 computers suddenly lose connectivity, the administrator may need to inspect the backbone cable.

In a mesh network, multiple links must be checked because there are many possible communication paths.

In a hybrid network, the administrator must identify which network segment is experiencing the problem and then troubleshoot the relevant topology.

Clarity of Presentation

Criteria	Star	Mesh	Bus	Hybrid
Maintenance	Easy	Difficult	Moderate	Moderate to Difficult
Fault Detection	Easy	Complex	Difficult	Moderate
Reliability	High	Very High	Low/Moderate	High
Troubleshooting	Simple	Complex	Difficult	Complex

Criteria	Star	Mesh	Bus	Hybrid
Cost	Moderate	High	Low	High
Technical Skill	Moderate	High	Low/Moderate	High
Failure Isolation	Easy	Very Good	Poor	Good